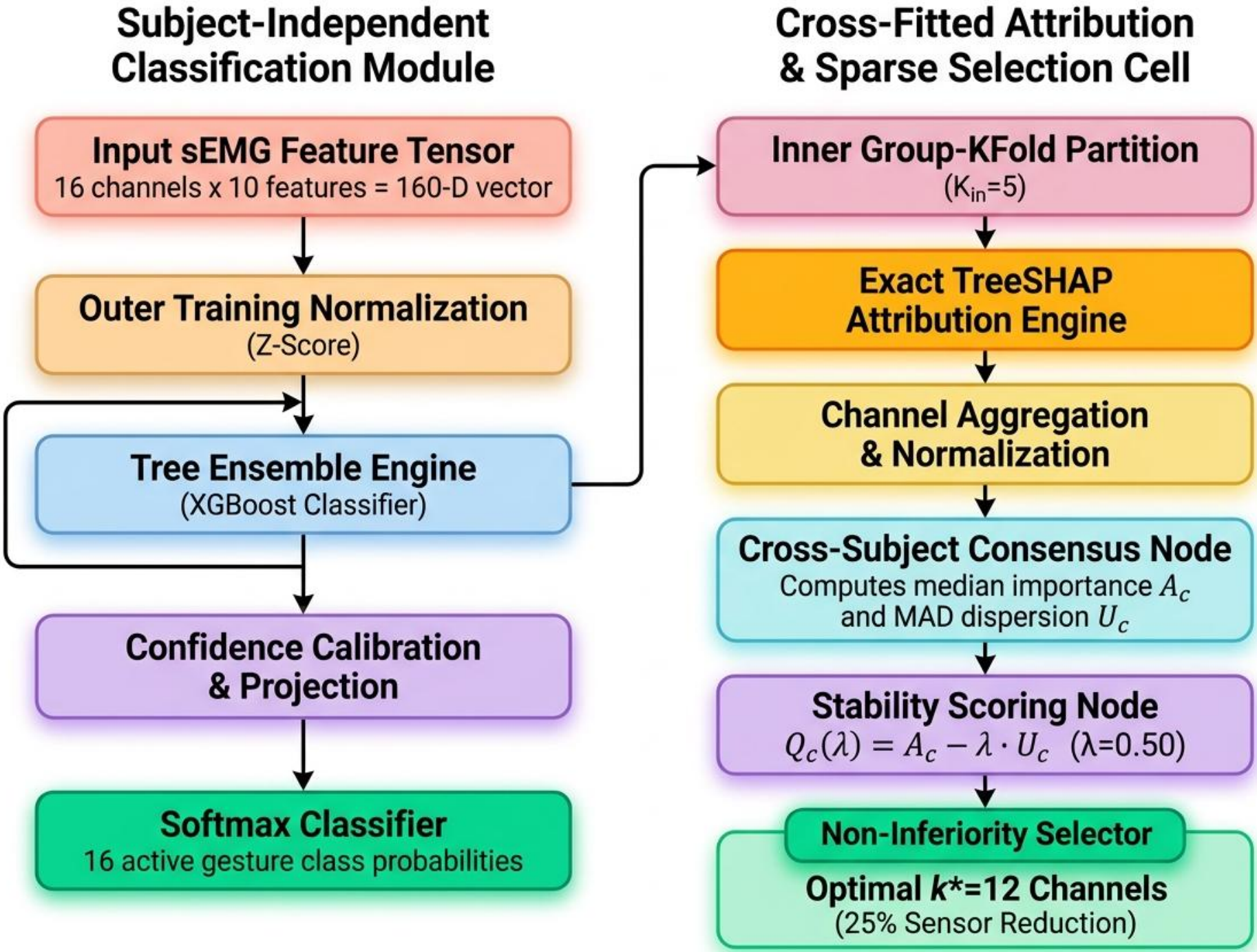

Subject-Independent Classification Module
Input sEMG Feature Tensor
16 channels x 10 features = 160-D vector
Outer Training Normalization
(Z-Score)
Tree Ensemble Engine
(XGBoost Classifier)
Confidence Calibration & Projection
Softmax Classifier
16 active gesture class probabilities
Cross-Fitted Attribution & Sparse Selection Cell
Inner Group-KFold Partition
(K_{in}=5)
Exact TreeSHAP Attribution Engine
Channel Aggregation & Normalization
Cross-Subject Consensus Node
Computes median importance A_c
and MAD dispersion U_c
Stability Scoring Node
$Q_c(\lambda) = A_c - \lambda \cdot U_c$ (λ=0.50)
Non-Inferiority Selector
Optimal k*=12 Channels
(25% Sensor Reduction)